

2024 Calculus II Additional Exercise (10)

1. Exercise 7, 12, 17 in page 40 from Calculus Supplement Homework.
2. Find the absolute extreme values of the function $w = xy + 2yz$ on the surface $x^2 + y^2 + z^2 - 10 = 0$.
3. Exercise 18 in page 40 from Calculus Supplement Homework.
4. Find $\iint_R x^2 e^{-y^2} dx dy$, where R is the triangle enclosed by the line $x = 0$, the line $y = 1$ and the line $y = x$.
5. Exercise 5 in page 46 from Calculus Supplement Homework.

Q1 P40 Ex 7

7. Assume $u(x, y)$ is twice continuously differentiable in a bounded closed region D with

$$\frac{\partial^2 u}{\partial x \partial y} \neq 0 \text{ and } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \text{ Then for } u(x, y),$$

- (A) both the maximum and minimum points are inside D .
- (B) both the maximum and minimum points are on the boundary of D .
- (C) the maximum point is inside D while the minimum point is on the boundary.
- (D) the minimum point is inside D while the maximum point on the boundary.

B

Reason: $\begin{vmatrix} \frac{\partial^2 u}{\partial x^2} & \frac{\partial^2 u}{\partial x \partial y} \\ \frac{\partial^2 u}{\partial y \partial x} & \frac{\partial^2 u}{\partial y^2} \end{vmatrix} < 0 \Rightarrow$ only can be saddle point.
inside D .

□

Ex 12

12. The absolute maximum of $u = \sin x \sin y \sin z$ with the restriction $x + y + z = \frac{\pi}{2}$ and $x, y, z > 0$ is ().

Sol: $F(x, y, z, \lambda) = \sin x \sin y \sin z + \lambda(x + y + z - \frac{\pi}{2})$

$$\begin{cases} 0 = \frac{\partial F}{\partial x} = \cos x \sin y \sin z + \lambda \\ 0 = \frac{\partial F}{\partial y} = \sin x \cos y \sin z + \lambda \\ 0 = \frac{\partial F}{\partial z} = \sin x \sin y \cos z + \lambda \\ 0 = \frac{\partial F}{\partial \lambda} = x + y + z - \frac{\pi}{2} \end{cases} \xRightarrow{\text{解得}} \begin{cases} \sin x, \sin y, \sin z \neq 0 (\because x, y, z > 0) \\ \sin(x-y) = \sin(y-z) = \sin(z-x) = 0 \\ x + y + z = \frac{\pi}{2} \end{cases}$$

$\xRightarrow{\text{解得}} x = y = z = \frac{\pi}{6} \Rightarrow \text{absolute maximum } u = (\sin \frac{\pi}{6})^3 = \frac{1}{8}.$

□

Ex 17

17. Suppose the tangent plane of the surface $z - e^z + 2xy = 3$ at P is $x + 2y = C$. Then $P = (\quad)$ and $C = (\quad)$.

$$P = (2, 1, 0), C = 4$$

$$\text{或 } P = (-2, -1, 0), C = -4$$

Sol: $F(x, y, z) = z - e^z + 2xy - 3$

$$\begin{aligned}\nabla F &= \frac{\partial F}{\partial x} \vec{i} + \frac{\partial F}{\partial y} \vec{j} + \frac{\partial F}{\partial z} \vec{k} \\ &= 2y \vec{i} + 2x \vec{j} + (1 - e^z) \vec{k}\end{aligned}$$

$$\therefore (2y \quad 2x \quad 1 - e^z)^T // (1 \quad 2 \quad 0)^T$$

$$\Rightarrow \begin{cases} x = 2y \\ z = 0 \end{cases} \xrightarrow{\text{代入 } F(x, y, z) = 0} 0 - 1 + 4y^2 - 3 = 0$$

$$\text{解得 } \begin{cases} x = \pm 2 \\ y = \pm 1 \\ z = 0 \end{cases} \Rightarrow P = (2, 1, 0) \text{ 或 } (-2, -1, 0)$$

$$\text{切平面: } 2(x-2) + 4(y-1) = 0 \text{ 或 } -2(x+2) - 4(y+1) = 0$$

$$\therefore C = 4 \text{ 或 } -4.$$

17

Q2

2. Find the absolute extreme values of the function $w = xy + 2yz$ on the surface $x^2 + y^2 + z^2 - 10 = 0$.

Sol: $F(x, y, z, \lambda) = xy + 2yz + \lambda(x^2 + y^2 + z^2 - 10)$

$$\begin{cases} 0 = \frac{\partial F}{\partial x} = y + 2\lambda x \end{cases}$$

① $\lambda = 0, y = 0, z = \pm\sqrt{2}, x = \pm 2\sqrt{2}$

$$\begin{cases} 0 = \frac{\partial F}{\partial y} = x + 2z + 2\lambda y \end{cases}$$

解得

② $\lambda \neq 0, \begin{cases} z = 2x \\ y = -\frac{5x}{2\lambda} = -2\lambda x \\ x^2 + y^2 + z^2 - 10 = 0 \end{cases}$

$$\begin{cases} 0 = \frac{\partial F}{\partial z} = 2y + 2\lambda z \end{cases}$$

$$\begin{cases} 0 = \frac{\partial F}{\partial \lambda} = x^2 + y^2 + z^2 - 10 \end{cases}$$

②.1 $x = 0, y = 0, z = 0$ 无解

②.2 $\lambda = \pm \frac{\sqrt{5}}{2},$

Then $x^2 + (2\lambda x)^2 + (2x)^2 = 10$
 $x^2 = 1$

$\therefore (x, y, z) = (1, \sqrt{5}, 2), (1, -\sqrt{5}, 2), (-1, -\sqrt{5}, -2), (-1, \sqrt{5}, -2)$

$\therefore w(-2\sqrt{2}, 0, \sqrt{2}) = 0$

$w(2\sqrt{2}, 0, -\sqrt{2}) = 0$

$w(1, -\sqrt{5}, 2) = -\sqrt{5} - 4\sqrt{5} = -5\sqrt{5}$

$w(-1, \sqrt{5}, -2) = -\sqrt{5} - 4\sqrt{5} = -5\sqrt{5}$

$w(1, \sqrt{5}, 2) = \sqrt{5} + 4\sqrt{5} = 5\sqrt{5}$

$w(-1, -\sqrt{5}, -2) = \sqrt{5} + 4\sqrt{5} = 5\sqrt{5}$

} absolute minimum value.

} absolute maximum value

Q3 P40 Ex 18

18. ★ The minimum of the distance between a point on $y = x^2$ and a point on $x - y - 2 = 0$ is ().

Sol: Assume $(u, v), (s, t)$ on curve respectively.

$$\text{Let } F(u, v, s, t, \lambda, \mu) = (u-s)^2 + (v-t)^2 + \lambda(u^2 - v) + \mu(s - t - 2)$$

$$\begin{cases} 0 = \frac{\partial F}{\partial u} = 2(u-s) + 2\lambda u & \text{(I)} \end{cases}$$

$$\begin{cases} 0 = \frac{\partial F}{\partial v} = 2(v-t) - \lambda & \text{(II)} \end{cases}$$

$$\begin{cases} 0 = \frac{\partial F}{\partial s} = -2(u-s) + \mu & \text{(III)} \end{cases}$$

$$\begin{cases} 0 = \frac{\partial F}{\partial t} = -2(v-t) - \mu & \text{(IV)} \end{cases}$$

$$\begin{cases} 0 = \frac{\partial F}{\partial \lambda} = u^2 - v & \text{(V)} \end{cases}$$

$$\begin{cases} 0 = \frac{\partial F}{\partial \mu} = s - t - 2 & \text{(VI)} \end{cases}$$

解得

$$\begin{cases} \mu + 2\lambda u = 0 \\ -\mu - \lambda = 0 \end{cases}$$

① $\lambda = 0$

Then $\begin{cases} v = t & \text{(V)} \\ u = s \\ \mu = 0 \end{cases} \Rightarrow s^2 - (s-2) = 0$ 无解

② $\lambda \neq 0 \Rightarrow u = \frac{1}{2}$

$\Rightarrow v = \frac{1}{4}$

$$\begin{cases} \frac{1}{2} - s + \frac{1}{4} - t = 0 \\ s - t - 2 = 0 \end{cases}$$

解得 $\begin{cases} s = \frac{11}{8} \\ t = -\frac{5}{8} \end{cases}$

$$\therefore d(u, v, s, t) = (u-s)^2 + (v-t)^2$$

$$d\left(\frac{1}{2}, \frac{1}{4}, \frac{11}{8}, -\frac{5}{8}\right) = \left(\frac{7}{8}\right)^2 + \left(\frac{7}{8}\right)^2 = 2 \cdot \left(\frac{7}{8}\right)^2 \Rightarrow \text{最小距离为 } \frac{7}{8}\sqrt{2}$$

□

Q4

4. Find $\iint_R x^2 e^{-y^2} dx dy$, where R is the triangle enclosed by the line $x = 0$, the line $y = 1$ and the line $y = x$.

Sol: 原式 = $\int_0^1 \left(\int_0^y x^2 e^{-y^2} dx \right) dy$

$$= \int_0^1 \frac{y^3}{3} e^{-y^2} dy$$

$$= \frac{1}{6} \int_0^1 y^2 e^{-y^2} dy^2$$

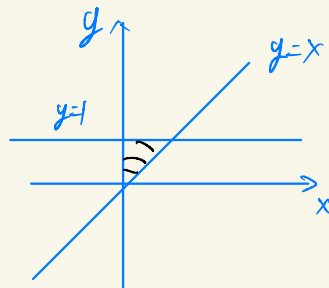
$$\stackrel{t=y^2}{=} \frac{1}{6} \int_0^1 t e^{-t} dt$$

$$= \frac{-1}{6} \int_0^1 t de^{-t}$$

$$= -\frac{1}{6} \left[t e^{-t} \Big|_0^1 - \int_0^1 e^{-t} dt \right]$$

$$= -\frac{1}{6} [e^{-1} + e^{-1} - 1]$$

$$= \frac{e-2}{6e}$$



D

Q5. P46 Ex5

5. Let $f(t) = \int_1^{t^2} e^{-x^2} dx$, then $\int_0^1 t f(t) dt = (\quad)$.

Sol: $f(t) = \int_1^{t^2} e^{-x^2} dx, f(1) = 0$

$$f'(t) = 2t e^{-t^4}$$

$$\text{Then } \int_0^1 t f(t) dt = \frac{1}{2} \left(t^2 f(t) \right) \Big|_0^1 - \int_0^1 t^2 f'(t) dt$$

$$= -\frac{1}{2} \int_0^1 2t^3 e^{-t^4} dt$$

$$= -\frac{1}{4} \int_0^1 e^{-t^4} dt^4$$

$s = t^4$

$$= -\frac{1}{4} \int_0^1 e^{-s} ds$$

$$= \frac{1}{4} e^{-s} \Big|_0^1$$

$$= \frac{1}{4e} - \frac{1}{4}$$